Experiment No. 9

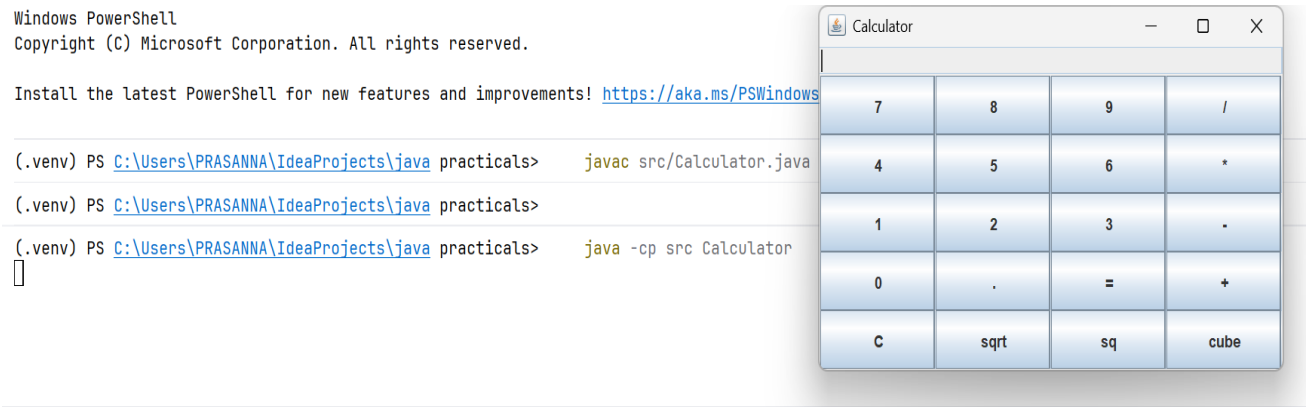
Aim : Designing the GUI using Swing components and IDE

1. Design a standard calculator using Swing components that supports basic operations
(Addition, Subtraction, Multiplication, and Division). Implement this with Intellij IDEA
Implementation Guidelines:
○ Use JTextField to display input/output.
○ Use JButton for digits (0-9) and operations (+, -, *, /, =, %, square, square-root,
cube, C, etc.).
○ Implement event handling for button clicks.
○ Display results in the text field

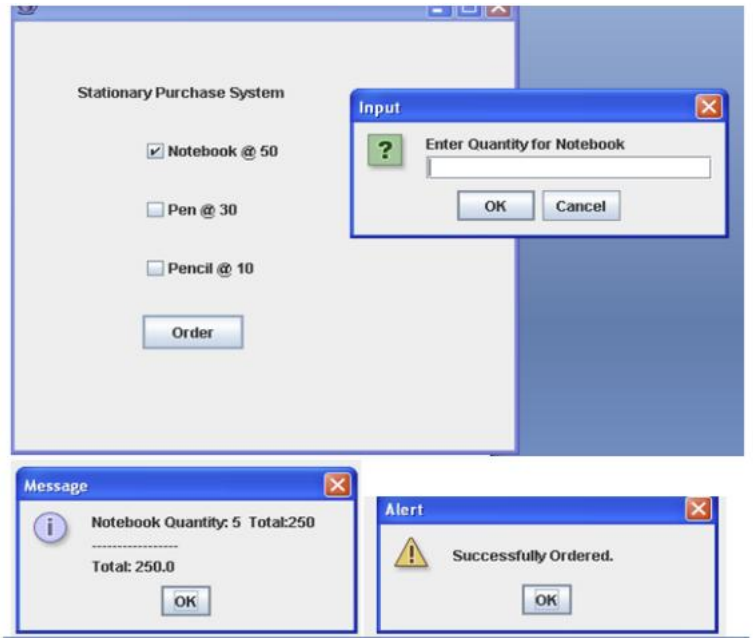
```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class Calculator extends JFrame implements ActionListener {
    private JTextField display;
    private String operator;
    private double firstOperand;

    public Calculator() {
        setTitle("Calculator");
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLayout(new BorderLayout());

        display = new JTextField();
        display.setEditable(false);
        add(display, BorderLayout.NORTH);

        JPanel buttonPanel = new JPanel(new GridLayout(5, 4));

        String[] buttonLabels = {
                "7", "8", "9", "/",
                "4", "5", "6", "*",
                "1", "2", "3", "-",
```

```
            "0", ".", "=", "+",
            "C", "sqrt", "sq", "cube"
        };

        for (String label : buttonLabels) {
            JButton button = new JButton(label);
            button.addActionListener(this);
            buttonPanel.add(button);
        }

        add(buttonPanel, BorderLayout.CENTER);
        pack();
        setLocationRelativeTo(null);
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        String command = e.getActionCommand();

        if (Character.isDigit(command.charAt(0)) || command.equals(".")) {
            display.setText(display.getText() + command);
        } else if (command.equals("C")) {
            display.setText("");
            operator = null;
            firstOperand = 0;
        } else if (command.equals("=")) {
            if (operator != null) {
                double secondOperand = Double.parseDouble(display.getText());
                double result = 0;
                switch (operator) {
                    case "+":
                        result = firstOperand + secondOperand;
                        break;
                    case "-":
                        result = firstOperand - secondOperand;
                        break;
                    case "*":
                        result = firstOperand * secondOperand;
                        break;
                    case "/":
                        result = firstOperand / secondOperand;
                        break;
                }
                display.setText(String.valueOf(result));
                operator = null;
            }
```

```
        } else if (command.equals("sqrt")) {
            double operand = Double.parseDouble(display.getText());
            display.setText(String.valueOf(Math.sqrt(operand)));
        } else if (command.equals("sq")) {
            double operand = Double.parseDouble(display.getText());
            display.setText(String.valueOf(operand * operand));
        } else if (command.equals("cube")) {
            double operand = Double.parseDouble(display.getText());
            display.setText(String.valueOf(operand * operand * operand));
        } else {
            operator = command;
            firstOperand = Double.parseDouble(display.getText());
            display.setText("");
        }
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> {
            new Calculator().setVisible(true);
        });
    }
}
```

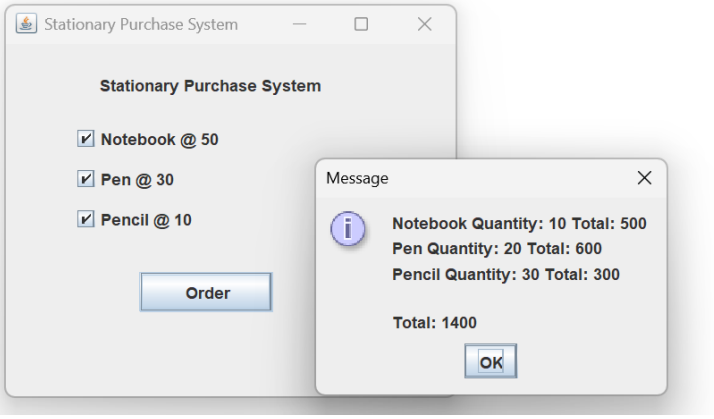
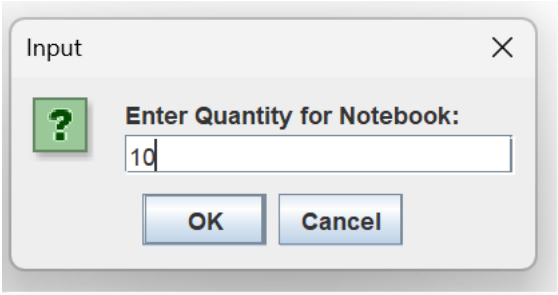
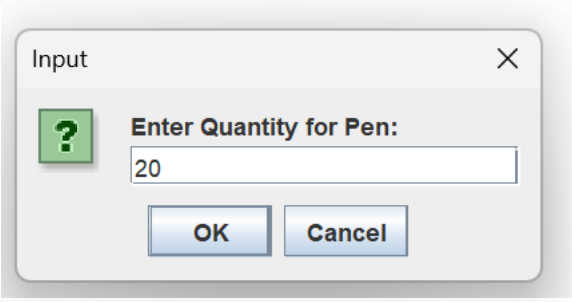
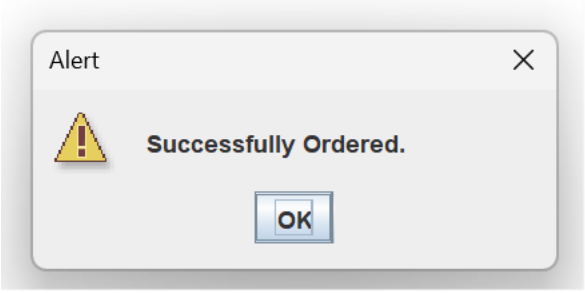
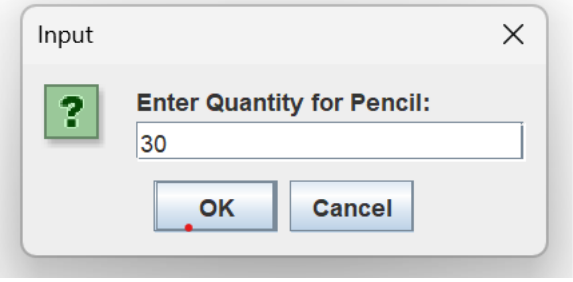


2. Implement the following problem statement using Intellij IDEA.




```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class StationaryPurchaseSystem extends JFrame implements ActionListener {

    JCheckBox notebook, pen, pencil;
    JButton orderBtn;

    int total = 0;

    public StationaryPurchaseSystem() {
        setTitle("Stationary Purchase System");
        setSize(350, 300);
        setLayout(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        JLabel title = new JLabel("Stationary Purchase System");
        title.setBounds(70, 20, 200, 20);
        add(title);

        notebook = new JCheckBox("Notebook @ 50");
        notebook.setBounds(50, 60, 150, 20);

        pen = new JCheckBox("Pen @ 30");
        pen.setBounds(50, 90, 150, 20);
```

```
        pencil = new JCheckBox("Pencil @ 10");
        pencil.setBounds(50, 120, 150, 20);

        add(notebook);
        add(pen);
        add(pencil);

        orderBtn = new JButton("Order");
        orderBtn.setBounds(100, 170, 100, 30);
        orderBtn.addActionListener(this);
        add(orderBtn);

        setVisible(true);
    }

    @Override
    public void actionPerformed(ActionEvent e) {

        total = 0;
        String message = "";

        if (notebook.isSelected()) {
            String qtyStr = JOptionPane.showInputDialog("Enter Quantity for Notebook:");
            if (qtyStr != null && !qtyStr.isEmpty()) {
                int qty = Integer.parseInt(qtyStr);
                int cost = qty * 50;
                total += cost;
                message += "Notebook Quantity: " + qty + " Total: " + cost + "\n";
            }
        }

        if (pen.isSelected()) {
            String qtyStr = JOptionPane.showInputDialog("Enter Quantity for Pen:");
            if (qtyStr != null && !qtyStr.isEmpty()) {
                int qty = Integer.parseInt(qtyStr);
                int cost = qty * 30;
                total += cost;
                message += "Pen Quantity: " + qty + " Total: " + cost + "\n";
            }
        }

        if (pencil.isSelected()) {
            String qtyStr = JOptionPane.showInputDialog("Enter Quantity for Pencil:");
            if (qtyStr != null && !qtyStr.isEmpty()) {
                int qty = Integer.parseInt(qtyStr);
```

```
            int cost = qty * 10;
            total += cost;
            message += "Pencil Quantity: " + qty + " Total: " + cost + "\n";
        }
    }

    message += "\nTotal: " + total;

    JOptionPane.showMessageDialog(this, message, "Message",
JOptionPane.INFORMATION_MESSAGE);
    JOptionPane.showMessageDialog(this, "Successfully Ordered.", "Alert",
JOptionPane.WARNING_MESSAGE);
  }

  public static void main(String[] args) {
    new StationaryPurchaseSystem();
  }
}
```

```
Terminal    Local
(.venv) PS C:\Users\PRASANNA\IdeaProjects\java practicals>    javac src/StationaryPurchaseSystem.java
(.venv) PS C:\Users\PRASANNA\IdeaProjects\java practicals>
(.venv) PS C:\Users\PRASANNA\IdeaProjects\java practicals>    java -cp src StationaryPurchaseSystem
```